

Continuous Assessment Test (CAT) – I – August 2026

Programme	:	2-year M.Tech (CSE, BDA, AI-ML, AI-DS)	Semester	:	Fall Semester 2026-2027
Course Code & Course Title	:	MACSE518 - Database Modeling and Design	Class Number	:	CH2026270103468
Faculty	:	Dr. R. Jothi	Slot	:	E1 + TE1
Duration	:	90 minutes	Max. Mark	:	50

General Instructions:

- Write only your registration number on the question paper in the box provided and do not write other information

Answer all questions

Q. No	Sub Sec.	Description	Marks	CO	BT Level
1		An e-commerce company maintains a centralized database that supports three applications: a Customer Mobile App, a Seller Portal, and a Warehouse Management System. To improve the system's performance, security, and scalability, the company redesigns its database architecture. Instead of allowing each application to access the database tables directly, customized database views are created for each application. Additionally, product images and inventory data are migrated to faster storage devices to enhance retrieval speed. The database schema is also modified by adding a new attribute, Product Rating, to the Product table. Furthermore, warehouse staff are restricted from viewing customer payment information to ensure that users can access only the data relevant to their roles. a) Explain how the Three-Schema Architecture helps improve security and maintainability in this e-commerce system. (6 Marks) b) For each of the changes mentioned above, identify whether it illustrates physical data independence or logical data independence. Justify your answer. (4 Marks)	10	CO1	K3
2		An electric vehicle (EV) charging company operates a network of charging stations across different cities. Each Charging Station is uniquely identified by a Station ID and stores its name, location, city, and operating hours. Every charging station contains several Charging Points, where each charging point is identified only by its Charging Point Number within that station (i.e., the same charging point number may exist in different stations). Every Charging Point	15	CO2	K3

		belongs to exactly one Charging Station and cannot exist independently. EV owners register with the company before using the charging service. Each EV Owner has a unique Owner ID, name, address, email, and one or more contact numbers. An owner may register multiple electric vehicles. Each Vehicle must be registered under exactly one EV Owner. Every Vehicle is uniquely identified by its Vehicle Registration Number and stores the manufacturer, model, battery capacity, and year of manufacture. Whenever a vehicle is charged, a Charging Session is created. A charging session records the Session ID, start time, end time, energy consumed (kWh), and total charging cost. Each charging session is associated with one vehicle and takes place at one charging point. A charging point can be used for many charging sessions over time. After every charging session, a Payment Receipt is generated. The Receipt Number is unique only within the corresponding charging session. Each receipt records the payment mode, transaction date, payment status, and amount paid. Every Payment Receipt is generated for exactly one Charging Session and cannot exist independently. - Draw an ER diagram for the above scenario. Clearly identify the following: - Strong entities and weak entities. (2 marks) - Primary keys for all entities. (2 marks) - Composite attributes and multivalued attributes, wherever appropriate. (2 mark) - Relationships between entities with appropriate cardinality ratios. (4 marks) - Convert the above ER model into a relational model by identifying the relations, primary keys, and foreign keys. (5 marks)			
3		Consider the following schema for an Airport Management System, where primary keys are underlined. PASSENGER (<u>PassengerID</u>, PassengerName, City) FLIGHT (<u>FlightID</u>, FlightName, Destination) BOOKING (<u>BookingID</u>, PassengerID, FlightID, TravelDate, TicketFare) - Write SQL queries for the following. (6 marks) - Display the Passenger Name, Flight Name, Destination, and Travel Date for all bookings.	15	CO3	K4

- ii. Display the names of passengers who have booked the flight 'Air India 621'.
 - iii. Display the destinations that have more than two bookings.
- b. Write the equivalent Relational Algebra expression for the following query.
- ```
SELECT P.PassengerName, F.FlightName FROM
PASSENGER P JOIN BOOKING B
ON P.PassengerID = B.PassengerID JOIN FLIGHT F
ON B.FlightID = F.FlightID WHERE F.Destination =
'Delhi';
```
- Draw an optimized query tree for the relational algebra expression that you have obtained. (5 Marks)
- c. Consider the following relations derived from the BOOKING and FLIGHT tables. Using the Relational Division, write the relational algebra expression to find the Passenger IDs of passengers who have travelled on every available flight. Also determine the final result of the expression using the given relations. (4 marks)

Bookings\_view (PassengerID, FlightID)

| Passenger ID | FlightID |
|--------------|----------|
| P101         | F101     |
| P101         | F102     |
| P102         | F101     |
| P103         | F101     |
| P103         | F102     |
| P103         | F103     |
| P104         | F102     |

Flights\_view(FlightID)

| FlightID |
|----------|
| F101     |
| F102     |
| F103     |

A railway reservation system maintains the following relation:  
 TRAIN (TrainNo, TrainName, Source, Destination, DepartureTime)

The relation contains approximately 2 million records and is stored in ascending order of TrainNo.  
 The following queries are frequently executed:

4

10

CO1

K4

- |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |  |  |  |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|
|  | <ul style="list-style-type: none"><li>• Search a train using TrainNo.</li><li>• Search all trains departing from Chennai.</li></ul> <p>a) Identify whether a Primary Index or a Secondary Index is more suitable for each of the above queries. Justify your answer. (3 Marks)</p> <p>b) If the file is no longer stored in the order of TrainNo, can a Primary Index still be used? Explain your answer. (2 Marks)</p> <p>c) Assume the system uses a B-tree of order 4 for indexing on TrainNo. Construct the B-tree by inserting the following keys in the given order:</p> <p>35, 20, 60, 10, 40, 80, 50, 30, 90</p> <p>Show the B-tree after each split and draw the final B-tree. (5 marks)</p> |  |  |  |
|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--|--|--|

\*\*\*\*\*All the best \*\*\*\*\*